

**REDUCED-ORDER PREDICTION OF OFFSHORE WIND TURBINE MOTION DURING
INSTALLATION USING GENERATIVE AI**

By

Dyllon Aiden Dunton

Bachelor of Computer Engineering, University of Maine Orono

Bachelor of Electrical Engineering, University of Maine Orono

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Advisory Committee:

Dr. Yifeng Zhu, Advisor

Dr. Amrit Verma

Dr. Andrew Goupee

Dr. Vikas Dhiman

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Dissertation Advisor: Dr. Yifeng Zhu

An Abstract of the Dissertation Presented
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This work develops a physics-informed latent-space learning framework for representing and predicting complex structural motion during offshore wind turbine blade installation. During installation, wave loading produces coupled horizontal tower-top motion with strong temporal and spectral structure. Although high-fidelity simulation can resolve this behavior, its computational cost limits its use in real-time prediction settings. The central goal of this work is therefore to develop a reduced-order machine learning framework that is both computationally efficient and physically consistent.

The proposed framework combines a physics-informed autoencoder with a latent diffusion predictive model. The autoencoder learns a compact latent representation of two-dimensional motion histories while preserving the dominant characteristics of the original signal. Physical consistency is encouraged through reduced-order governing equations identified from OrcaFlex simulation data using Sparse Identification of Nonlinear Dynamical Systems (SINDy) and enforced as a soft constraint during training. Future motion is modeled using a conditional latent diffusion model driven by prior motion and wave-state information. The predictive framework is formulated to support efficient latent-space forecasting while maintaining compatibility with the learned physical structure of the motion data.

The framework is evaluated using tower-top motion data generated in OrcaFlex across varying wave conditions. Results show that the learned latent representation preserves key motion characteristics while achieving low reconstruction error and supporting future-state prediction in a reduced-order setting. More broadly, this work establishes a foundation for physics-guided diffusion-based predictive models that can support real-time digital twin applications in marine operations.

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Keywords: latent diffusion, physics-informed learning, autoencoder, real-time prediction, offshore wind turbine installation

This work develops a machine learning framework for representing and predicting the motion of an offshore wind turbine during blade installation. At this stage of installation, ocean waves cause the top of the turbine structure to move in complicated horizontal patterns. Predicting this motion is important because it affects how safely and efficiently installation can be performed. Although detailed simulation tools can model this behavior, they are too computationally expensive for real-time use.

The proposed framework uses two connected machine learning components. The first is an autoencoder, which compresses motion data into a much smaller representation while preserving its most important features. To keep this representation physically meaningful, the model is guided by simplified governing equations identified from OrcaFlex simulation data using a method called SINDy. The second component is a latent diffusion predictive model, which uses the compact representation to forecast future motion from prior motion and wave conditions.

The framework is evaluated on simulated turbine-motion data generated across a range of wave conditions. Results show that the method preserves key motion behavior, achieves low reconstruction error, and supports reduced-order prediction of future motion. More broadly, this work supports the development of real-time digital twin tools for offshore engineering applications.

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Chapter 1

INTRODUCTION

Problem Statement

Offshore wind turbine installation is a complex marine operation carried out under uncertain environmental conditions. Among the installation stages, blade mating is one of the most challenging because it requires precise alignment while the partially assembled structure is undergoing wave-induced motion. In the hammerhead configuration, the monopile, transition piece, tower, and nacelle are installed before blade attachment, creating a temporary structural arrangement with an eccentric mass. Under these conditions, the tower-top response exhibits coupled motion in the fore-aft and side-side directions. Rather than moving independently along two axes, the tower top can trace orbital trajectories whose amplitude, orientation, and stability depend strongly on the incident wave conditions, as shown in Figure 1.1.

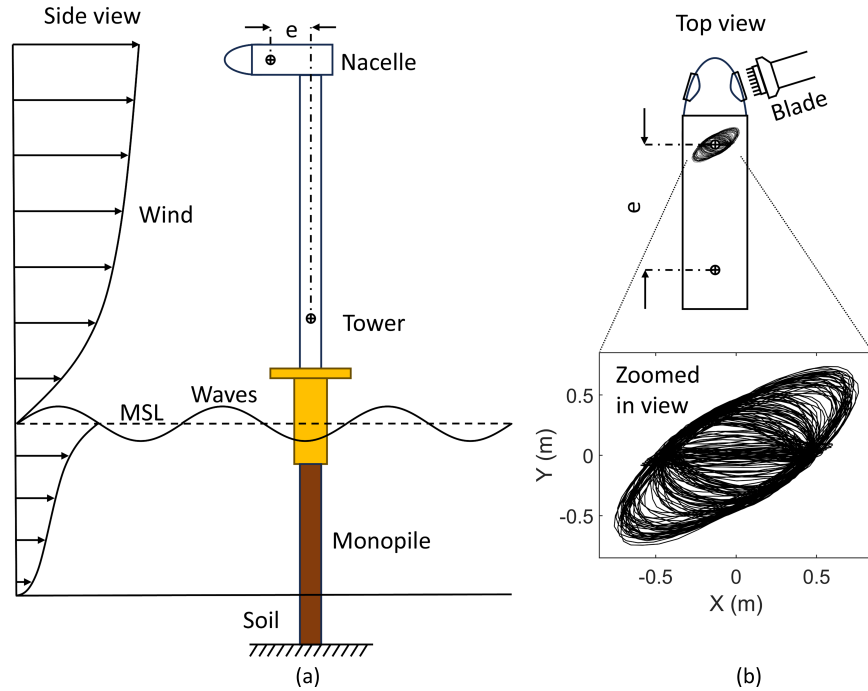


Figure 1.1 Hammerhead configuration of an offshore wind turbine during blade installation. (a) Side view with eccentric nacelle offset; (b) top view showing tower-top orbital trajectories.